

Lab 4: Vibe Coding — Chat History & Prompt Transcript

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Course: Software Engineering Lab (UE24CS252AA2)	Date: October 9, 2026
Assigned Repo: SETAPESU26/50_traffic-escape	Personal Repo: dhruv5112/50_traffic-escape
Vibe Tool: ChatGPT / OpenAI Vibe Coding Agent	Target Objective: 3 to 4 prompt iterations

Prompt 1 (Bug Inspection & Resolution) — Dhruv U Rajesh:

I have cloned the assigned repository SETAPESU26/50_traffic-escape. Upon inspecting and running python main.py, I recorded a 10-second gameplay video showing broken behavior. Specifically, I identified three bugs:

1. The player can move down continuously past the bottom sidewalk completely off the screen.
2. Cars entering or leaving the screen can hit the player while still on the bottom safe sidewalk.
3. Fractional car speeds are truncated because position updates use integer rounding incorrectly.

Please fix these bugs cleanly in player.py, game_engine.py, and traffic.py while preserving the original core game mechanics.

AI Assistant:

I inspected the codebase and addressed all three bugs:

- **Off-screen bottom movement:** In game/player.py, updated move() to clamp vertical movement with `max_y=600` so `ny = max(0, min(max_y - self.rect.height, self.rect.y + dy))`, preventing the car from descending past the lower sidewalk.
- **Sidewalk collision safety:** In game/game_engine.py, added a boundary guard so collisions are only evaluated when `self.player.rect.bottom < HEIGHT - 50`, protecting the player while resting on the bottom sidewalk.
- **Fractional speeds:** In game/traffic.py, added a floating-point tracker `self.y` in Car, updating `self.y += self.direction * self.speed` and synchronizing `self.rect.y = round(self.y)`.

Tested the fixes; the player now respects all boundaries and the safe sidewalk works as expected.

Prompt 2 (Task 1: Lives System & Task 2: Moving Raft Lane) — Dhruv U Rajesh:

Now let's implement the first two features from the repository README:

- **Task 1 (Lives System):** Give the player 3 lives. Colliding with a car should decrement 1 life, trigger a safe respawn with 1.5 seconds of flashing invulnerability, and only trigger Game Over when lives reach 0.
 - **Task 2 (Moving Log/Raft Lane):** Add a safe river strip in the middle of the road where the player must hop onto a moving raft to cross. The raft should carry the player horizontally along with the current. Stepping into water or drifting off-screen must cost a life. Traffic should not collide with the player in this river strip.
- Please implement these features modularly.

AI Assistant:

Implemented Task 1 and Task 2:

- **Task 1 (Lives System):** Initialized `self.lives = 3` and `self.invulnerable = 0` in `GameEngine`. Added `lose_life()` method: when hit, if `lives > 0`, decrements life, respawns player at start coordinates (`WIDTH//2, HEIGHT-80`), and sets `self.invulnerable = 90` frames (1.5 seconds) with alternating flash rendering in `draw()`. Game over triggers only at `lives == 0`.
- **Task 2 (Moving Rafts):** Created `game/raft.py` defining `RIVER_TOP=120` and `RIVER_BOTTOM=210`, rendering a blue river channel and animated floating wooden log rafts. In `update()`, when the player enters the river zone, we verify their center overlaps an active raft: if yes, `self.raft_x` carries the player with the raft's velocity; if they fall in open water or drift off screen, `lose_life()` is triggered. Traffic collisions are disabled inside the river strip. Updated the HUD to display current remaining lives.

Prompt 3 (Task 3: High Score Persistence & Task 4: Day/Night Cycle) — Dhruv U Rajesh:

Next, let's implement the remaining two tasks from the README:

- **Task 3 (High Score Table):** Track and display the top 5 scores across game sessions, persisted to `high_scores.json`. Use atomic writing to guarantee file integrity without corruption, and render the scores on the HUD.
- **Task 4 (Day/Night Cycle):** Switch between Day and Night every 30 seconds of active gameplay. When night falls, darken the background and give oncoming cars illuminated directional headlight beams casting 170 pixels ahead. Ensure all edge cases (such as restarting or corrupted JSON files) are handled cleanly.

AI Assistant:

Implemented Task 3 and Task 4:

- **Task 3 (High Score Persistence):** Created `game/high_scores.py` with `HighScores` class. Loads from `high_scores.json` with safe error fallback for missing/malformed JSON. The `add(score)` method sorts descending, retains top 5, and writes atomically using a temporary file (`.tmp`) followed by an atomic replacement (`replace()`). Renders TOP 5: ... on the lower display bar.
 - **Task 4 (Day/Night Cycle):** In `game_engine.py`, tracking `self.elapsed_seconds += dt` and calculating `self.night = int(elapsed // 30) % 2 == 1`. In `draw()`, background switches between day (`60,60,60`) and night (`24,27,42`). In `game/traffic.py`, added alpha-blended polygon headlight beams extending 170px ahead of each car in its travel direction with illuminated headlamp circles.
- Both systems tested and verified across restarts.

Prompt 4 (Automated Tests, Verification, README Checklist & Packaging) — Dhruv U Rajesh:

To finalize the lab deliverables:

1. Write an automated test suite in `tests/test_game.py` verifying boundary checks, life deductions, raft riding, night cycle transitions, and high score JSON persistence.
2. Check off all items in the `README.md` submission checklist and document folder structure.
3. Record 10-second before/after gameplay MP4 videos.
4. Organize all changes into separate git commits per task and push to my personal GitHub repository `dhruv5112/50_traffic-escape`.

AI Assistant:

Verification and submission packaging complete:

- **Unit Tests:** Created tests/test_game.py with 7 automated tests covering boundary enforcement, 3-hit game over, respawn flashing invulnerability, raft carrying and water drowning, high score persistence & invalid JSON resilience, and day/night cycle timing. All 7 tests passed (OK).
- **README Checklist:** Updated README.md marking all submission items checked ([x]) and detailing the complete folder tree.
- **Gameplay Recordings:** Rendered videos/before.mp4 (demonstrating off-screen movement and instant crash) and videos/after.mp4 (demonstrating boundary fix, 3 lives, raft navigation, night mode with headlights, and finish) — both exactly 10.00 seconds at 30 FPS.
- **Git History:** Committed all tasks as separate, well-documented commits under your GitHub identity dhruv5112 <dhruv280306@gmail.com> and pushed to origin main on github.com/dhruv5112/50_traffic-escape.